

Kaixin Zheng

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EDUCATION

University of Waterloo

Waterloo, ON, Canada

Doctor of Philosophy in Computer Science

Sept. 2025–present

- Supervisor: Dr. Anita Layton.
- GPA: 92.5/100 (equivalent to 4.0/4.0).

Master of Mathematics in Applied Mathematics (Thesis-based)

Jan. 2023–Jan. 2025

- Supervisor: Dr. Anita Layton.
- GPA: 92.5/100 (equivalent to 4.0/4.0).
- Thesis: “Mathematical Models of Kidney Function: Effects of Hypertension and Circadian Rhythm.”

New York University

New York, NY, United States

Bachelor of Science in Honors Mathematics; Minor in Computer Science

Sept. 2019–May 2022

- GPA: 3.89/4.0 (summa cum laude, top 5% across all majors).

University of California, Irvine

Irvine, CA, United States

Bachelor of Science in Physics; transferred to New York University

Sept. 2018–June 2019

- GPA: 4.0/4.0.

HONORS AND AWARDS

Women in Mathematics Mentorship Award, University of Waterloo

Apr. 2025

Graduate Research Studentship, University of Waterloo

Sept. 2025–June 2029

- Departmental research funding covering tuition and stipend for PhD studies.

International Doctoral Student Award, University of Waterloo

Sept. 2025–June 2029

- Multi-year award (~CAD 15,420/year) for international PhD students.

Graduate Research Studentship, University of Waterloo

Jan. 2023–Jan. 2025

- Departmental research funding covering full tuition and partial stipend for research-based Master of Mathematics studies.

Dean’s List, New York University

Fall 2019–Spring 2022

Dean’s Honor List, University of California, Irvine

Fall 2018–Spring 2019

RESEARCH EXPERIENCE

Graduate Researcher

Jan. 2025–present

David R. Cheriton School of Computer Science, University of Waterloo

Waterloo, ON, Canada

Supervisor: Dr. Anita Layton

- Developing scientific machine learning methods for surrogate modeling of dynamical systems.
- Investigating inverse problems, with a focus on parameter estimation in scientific dynamical systems.
- Deriving mathematical guarantees for proposed methods and analyzing failure modes of existing approaches.

Graduate Researcher

Jan. 2023–Jan. 2025

Applied Mathematics Department, University of Waterloo

Waterloo, ON, Canada

Supervisor: Dr. Anita Layton

- Applied numerical methods to model mammalian kidney function.
- Studied sex- and circadian-dependent effects on drug efficacy in hypertension.
- Developed nephron and renal circadian models to predict solute/water transport and analyze drug-timing effects.

Undergraduate Research Assistant

Courant Institute of Mathematics, New York University

Advisor: Dr. David McLaughlin

June 2021–Dec. 2021

New York, NY, United States

- Reviewed literature on calcium diffusion and PDE-based modeling in dendritic spines.
- Studied basic dendritic spine models and analyzed calcium dynamics.

PUBLICATIONS

Peer-reviewed

- Zheng, K., & Layton, A. T. (2026). Circadian rhythms in glomerular filtration govern natriuresis and diuretic responsiveness. *Frontiers in Physiology*, 17, 1828410. <https://doi.org/10.3389/fphys.2026.1828410>
- Zheng, K., & Layton, A. T. (2024). Predicting sex differences in the effects of diuretics in renal epithelial transport during angiotensin II-induced hypertension. *American Journal of Physiology–Renal Physiology*, 326(5), F737–F750. <https://doi.org/10.1152/ajprenal.00398.2023>

Book Review

- Abo, S., Cheung, C. C., Dasgupta, R., Dutta, P., Hakimi, S., Kaur, A., . . . & Zheng, K. (2023). Featured Review: Mathematical Modeling for Epidemiology and Ecology.

COURSE PROJECTS

Project reports are available on my <https://kaixin-zheng.github.io/projects/>.

Polynomial Auditing for Black-Box DP Mechanisms

Winter 2026

CS 860: Algorithms for Private Data Analysis

University of Waterloo

- Proposed a polynomial upper-envelope framework for black-box auditing of differential privacy.
- Converted fitted trade-off curves into audited (ϵ, δ) guarantees via tangent geometry.
- Evaluated the method on Gaussian mechanisms and toy DP-SGD.

Modeling Drug Resistance in Bladder Cancer

Winter 2024

AMATH 881: Introduction to Mathematical Oncology

University of Waterloo

- Modeled spontaneous, drug-induced, and CSCs-related drug resistance in bladder cancer.
- Compared dosage strategies under the same cumulative dosage.

Lab Book on Asymptotic Analysis and Perturbation Theory

Fall 2023

AMATH 732: Asymptotic Analysis and Perturbation Theory

University of Waterloo

- Created a lab book on asymptotic expansions, perturbation methods, and signal processing.
- Provided proofs and MATLAB visualizations of normal modes in N -coupled oscillators.

TEACHING

Graduate Teaching Assistant

David R. Cheriton School of Computer Science, University of Waterloo

Courses:

CS 371 Intro to Computational Mathematics

CS 335 Computational Methods in Business and Finance

Graduate Teaching Assistant

Applied Mathematics Department, University of Waterloo

Courses:

SYDE 211 Calculus 3

MATH 137 Honors Calculus 1

MATH 237 Advanced Calculus 3

AMATH 331 Applied Real Analysis

AMATH 382 Computational Modelling of Cellular Systems

Grader

Courant Institute of Mathematical Sciences, New York University

Courses:

MATH-UA.251 Intro to Mathematical Modeling

MA-UY.4414 Applied Partial Differential Equations

Sept. 2021–Dec. 2021

New York, NY, United States

SKILLS

Research Areas: Scientific machine learning, Computational math, Computational physiology

Domain Expertise: Renal physiology, Mechanistic modeling, Sex differences/circadian regulation in drug response

Machine Learning: Operator learning, Statistical learning theory, Privacy-preserving ML

Mathematics Skills: ODE/PDE, Dynamical systems, Probability

Programming: Python (PyTorch, NumPy, SciPy), MATLAB

MENTORSHIP

Undergraduate Research Mentor

Faculty of Mathematics, University of Waterloo

Jan. 2025–Apr. 2025

Waterloo, ON, Canada

VOLUNTEER

Student Representative at Math Grad Visit Day

Applied Mathematics Department, University of Waterloo

Mar. 2024

Waterloo, ON, Canada